

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 3 – CLASS TEST (5 QUESTIONS)

Course Code:	MLA0407	Course Title:	Deep Learning
CO Assessed:	CO2	Assessment:	Assessment Tool 3 – CLASS TEST
Weightage:	35% of CIA	Total Marks:	(5 Questions × 10 Marks)
CO1 Statement:	<i>Apply the fundamental concepts of machine learning and neural networks to design and analyze learning algorithms using perceptrons, multilayer perceptrons, forward propagation, backpropagation, gradient descent optimization techniques, and Maximum Likelihood Estimation for solving real-world problems.</i>		
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Section / Batch:	2024	Date:	11-08-2026

Code of Conduct

I S.Venkateswarlu/192425340 (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

QUESTIONS: 5

Total Marks: 50

Annexure A

Class Test

Duration: 1hr

1. A company developing an AI-based disease diagnosis system is deciding whether to use a traditional Machine Learning model or a Deep Learning model. Compare both approaches by discussing their feature extraction methods, data requirements, computational needs, and suitability for image-based medical diagnosis.
2. A facial recognition application receives raw image data without any manually extracted features. Explain how Representation Learning enables the model to automatically learn useful features from the images and discuss its advantages over manual feature engineering.
3. Two neural network models are designed for an image classification task. One model contains many hidden layers with fewer neurons, while the other has a single hidden layer with a large number of neurons. Compare these architectures and discuss how width and depth influence learning performance.
4. A Deep Neural Network experiences slow learning due to the vanishing gradient problem during training. Compare the Sigmoid and Tanh activation functions and explain their impact on gradient flow, convergence speed, and overall network performance.
5. An image classification model suffers from inactive neurons during training, leading to poor prediction accuracy. Evaluate the use of ReLU, Leaky ReLU, and ELU activation functions for this problem and recommend the most suitable choice with justification.
6. A multiclass image classification system is designed to classify animals into several categories such as cats, dogs, birds, and horses. Explain how the Softmax activation function processes the final output layer and converts network outputs into class probabilities.
7. A company collects millions of unlabeled customer behavior records and wants to discover hidden patterns without assigning labels to the data. Discuss how unsupervised training of neural networks can be applied to this problem and explain its advantages in pattern discovery.

8. A music streaming platform aims to improve song recommendations by identifying hidden relationships between users and songs from historical listening data. Explain how a Restricted Boltzmann Machine (RBM) can be used in this application by describing its architecture and learning process.

9. An industrial organization wants to reduce storage requirements for large sensor datasets while preserving important information. Explain how an Autoencoder performs data compression and reconstruction, and describe the role of the encoder, bottleneck layer, and decoder.

10. A smart city project plans to implement Deep Learning for applications such as face recognition, autonomous vehicles, speech recognition, and medical image analysis. Discuss how Deep Learning improves the efficiency and accuracy of these applications compared to conventional approaches.

1. Traditional Machine Learning vs. Deep Learning for Disease Diagnosis

When developing an AI-based disease diagnosis system (such as identifying tumors in MRI scans), the architectural choice fundamentally dictates the engineering pipeline.

- **Feature Extraction:** Traditional Machine Learning (ML) relies heavily on manual feature engineering. Domain experts must hand-craft algorithms to identify textures, shapes, or pixel intensity histograms (e.g., using SIFT or Gabor filters). Deep Learning (DL), specifically Convolutional Neural Networks (CNNs), utilizes representation learning to automatically extract hierarchical features directly from raw pixel arrays—learning simple edges in early layers and complex biological structures in deeper layers.
- **Data Requirements:** ML models can converge effectively on smaller, well-curated datasets. DL models are parameter-heavy and require massive amounts of annotated medical data to generalize effectively and avoid severe overfitting.
- **Computational Needs:** Traditional ML algorithms can typically be trained on standard CPUs with lower memory requirements. DL demands high-performance parallel computing (GPUs or TPUs) to handle millions of parameters and intensive matrix multiplications during backpropagation.
- **Suitability for Image-Based Diagnosis:** Deep Learning is unequivocally superior for this task. Medical images contain intricate, highly abstract, and non-linear spatial patterns that are nearly impossible to fully capture through human-engineered rules. DL's ability to autonomously map these complex hierarchical structures makes it the industry standard for computer vision in healthcare.

2. Representation Learning in Facial Recognition

In a facial recognition system, representation learning allows the neural network to bypass human-defined rules (like measuring the distance between eyes) and learn the optimal representations directly from raw, unannotated pixel data.

- **How it Works:** As raw image data passes through the network, the model autonomously applies transformations. Early convolutional layers act as edge detectors by capturing local pixel gradients. Middle layers synthesize these edges into distinct facial motifs (e.g., a nose or an eyebrow). The final layers flatten this data into a highly compressed, low-dimensional mathematical vector (an embedding) that uniquely represents a specific identity.
- **Advantages Over Manual Feature Engineering:**
 - **Removes Human Bias:** The model is not restricted by what a human engineer *thinks* is important for identifying a face.
 - **Robustness to Variance:** It naturally learns to account for variations in lighting, facial hair, aging, and angles that would break rigid, hand-coded algorithmic rules.
 - **Scalability:** It scales effortlessly to millions of identities, whereas manually engineering features for every new demographic or edge case is computationally and practically impossible.

3. Neural Network Architecture: Width vs. Depth

When designing an image classification network, the distribution of neurons dictates the model's capacity to learn abstract concepts.

- **Shallow and Wide (Single hidden layer, high neuron count):** Based on the Universal Approximation Theorem, a sufficiently wide hidden layer can theoretically approximate any continuous function. However, this architecture is highly inefficient for image classification. It forces the network to memorize local spatial mappings rather than understanding underlying patterns, making it extremely prone to overfitting and parameter bloat.
- **Deep and Narrow (Many hidden layers, fewer neurons):** Depth fosters **feature compositionality**. By stacking layers, the network learns to build complex concepts out of simpler ones (pixels $\rightarrow$ edges $\rightarrow$ textures $\rightarrow$ objects).

- **Influence on Learning Performance:** Depth exponentially increases the expressive power and generalization capabilities of the model for complex data like images. Deep networks can model highly non-linear decision boundaries with significantly fewer total parameters than a wide, shallow network attempting the same task.

4. The Vanishing Gradient Problem: Sigmoid vs. Tanh

The vanishing gradient problem occurs during backpropagation when the derivatives of the activation function are very small. As these small gradients are multiplied backward through multiple layers, they shrink exponentially, preventing early layers from updating their weights.

- **Sigmoid Activation Function:** Mathematically defined as $\sigma(x) = \frac{1}{1 + e^{-x}}$. It maps inputs to a range of $[0, 1]$.
 - **Impact:** The maximum derivative of a Sigmoid function is only 0.25. When chained together, gradients vanish rapidly. Furthermore, the output is strictly positive (not zero-centered), which forces all weight gradients in a layer to move in the same direction, causing erratic and slow convergence during optimization.
- **Tanh Activation Function:** Mathematically defined as $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$. It maps inputs to a range of $[-1, 1]$.
 - **Impact:** Tanh has a maximum derivative of 1, which results in stronger gradients compared to Sigmoid. Because its output is zero-centered, it ensures well-balanced gradient updates and significantly faster convergence.
- **Overall Network Performance:** While Tanh mitigates the vanishing gradient better than Sigmoid and is the superior choice for hidden layers, both functions suffer from "saturation" at extreme positive or negative inputs. Consequently, neither is recommended for very deep neural networks.

5. Inactive Neurons (Dying ReLU): Evaluation and Recommendation

The "Dying ReLU" problem happens when a large gradient update causes a neuron's weights to shift so drastically that it constantly outputs negative values. Because the derivative of ReLU for negative inputs is exactly zero, the neuron can no longer update its weights and becomes permanently "dead," crippling the network's predictive accuracy. **Evaluation of Activation Functions:**

- **ReLU (Rectified Linear Unit):** $f(x) = \max(0, x)$. While computationally efficient and excellent for positive values, it strictly zeroes out negative inputs, making it the direct cause of the dying neuron problem.

- **Leaky ReLU:** $f(x) = \max(\alpha x, x)$ where α is a small constant (e.g., 0.01). By introducing a slight slope for negative inputs, it ensures the gradient never hits absolute zero, allowing "dead" neurons a chance to recover and continue learning.
- **ELU (Exponential Linear Unit):** $f(x) = x$ for $x > 0$, and $\alpha(e^x - 1)$ for $x \leq 0$. ELU provides a smooth, non-zero gradient for negative values, preventing dying neurons while also pushing the mean of all activations closer to zero. This zero-centering drastically speeds up learning and improves generalization.

Recommendation and Justification:

For a network suffering from severe neuron death and poor accuracy, **ELU is the most suitable choice**.

- **Justification:** ELU mathematically solves the dead neuron problem while simultaneously stabilizing the training process through zero-centered outputs. If computational resources are strictly limited (as ELU requires calculating expensive exponential functions), **Leaky ReLU** serves as a highly effective, computationally cheap secondary recommendation to force gradient flow and restore network capacity.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, wellanalyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____